

LoRaWAN Research Test Manual

This manual contains only the Chapter 3 tests, measurements, acceptance rules, and released operator commands.

chapters/Zacarias_Chapter3.pdf controls **what must be tested and measured**. Verified live infrastructure controls **how the test is executed**. Stale architecture details from Chapter 3 are not used as implementation instructions.

Rules used for every test

- Keep the raw evidence used to calculate every result.
- Keep the numerator and denominator for every percentage.
- Keep the start and end timestamps for every latency and recovery measurement.
- Report time in ms or s, bandwidth in Mbps, transaction/block size in bytes, throughput in records/s or TPS, CPU and memory in %, and counts as integers.
- Keep gateway and server resource measurements separate.
- Grafana and the research cockpit are for live observation. Final values come from the recorded run evidence.
- PASS means the intended condition occurred and the expected result was proven.
- FAIL means the intended condition occurred but the result was wrong.
- INVALID means the intended condition or required evidence was not achieved; retain the attempt and repeat it with a new ID.
- BLOCKED means a required fixture, authorization, or methodology is unavailable before the test starts.
- A code block appears in this manual only when the complete test action is currently READY. If a test is not ready, its measurements stay here but no placeholder code is shown.

Test status

Test	Required work	Current operator status
PRE	Full technical preflight	READY
P1	Normal-operation performance	CAPTURE READY; live sensor transmission boundary not proven
P2	Fabric performance and overhead	HARNESS REQUIRED
R1	Internet interruption and recovery	CAPTURE READY; outage orchestrator not released
R2	Fabric unavailability and reconciliation	HARNESS REQUIRED
A1	Authentication and access control	CAPTURE READY; complete 90-trial orchestrator not released
A2	Endorsement-policy enforcement	HARNESS REQUIRED
S1	LoRaWAN replay and spoofing	CAPTURE READY; complete attack orchestrator not released
S2	Application duplicate/replay	METHODOLOGY REQUIRED
F1	MQTT invalid-connection flooding	CAPTURE READY; isolated-listener lifecycle not released
F2	Invalid application-message flooding	CAPTURE READY; isolated validation fixture not released
I1	Application-layer alteration	HARNESS REQUIRED
I2	Post-storage database tampering	HARNESS REQUIRED

Test	Required work	Current operator status
I3	Blockchain duplicate/overwrite	HARNESS REQUIRED
T1	Individual record traceability	CAPTURE READY; complete scoring orchestrator not released
T2	Five-record history reconstruction	CAPTURE READY; complete scoring orchestrator not released

PRE - Full technical preflight

Run this immediately before a formal test group. It checks the current recorder/tooling state, counted firmware archive, workstation/cloud/gateway timing, required sensor USB identities, database role, evidence-service readiness, gateway LTE production route, and a recent full-stack control.

```
python .\test\automation\research-manual\run_test.py PRE
```

When prompted, type EXECUTE. Do not start counted work if PRE fails.

P1 - Normal-operation performance

Design

- 3 runs.
- 30 minutes per run.
- EMU-01 sends one legitimate uplink every 15 s.
- About 120 scheduled transmissions per run.
- No intentional attack, outage, flood, or service failure.

Measure

- Packet-delivery rate: $\text{unique legitimate uplinks accepted by ChirpStack} / \text{scheduled EMU-01 transmission attempts} \times 100$.
- End-to-end latency for each matched reading: $T_{\text{database_storage}} - T_{\text{sensor_transmission}}$ when the clock/correlation gate proves that boundary is valid.
- Mean, sample SD, minimum, and maximum latency.
- System throughput: $\text{unique legitimate readings stored} / \text{test duration}$.
- Fabric transaction success rate: $\text{valid committed} / \text{submitted} \times 100$.
- Gateway CPU average and maximum.
- Gateway memory average and maximum.
- Server CPU average and maximum per participating server.
- Server memory average and maximum per participating server.

Required evidence

EMU-01 source transmissions, ChirpStack acceptance, database rows, Fabric/outbox result, resource samples, run timestamps, and the sealed run checksum manifest.

Operator block

Not released. The recorder/finalization path is proven, but the live publication rehearsal captured zero EMU-01 SENSOR_TX attempts and the gateway reported zero received RF packets. Re-release the command only after a clean rehearsal produces source attempts and a measurable ChirpStack PDR boundary.

P2 - Fabric performance and overhead

Design

- Valid workloads: 1 transaction every 15 s, 1 TPS, 5 TPS, and 10 TPS.
- 5 minutes per workload.
- 3 repetitions per workload = 12 with-Fabric workload runs.
- Compare the same processing workload with and without Fabric.
- Chapter 3 does not prescribe the number of no-Fabric comparison runs. Freeze the comparison design before counted execution and do not select it after seeing results.

Measure

- Commit latency per transaction: $T_{\text{commit}} - T_{\text{submission}}$.
- Mean and sample SD of commit latency.
- Transaction throughput: $\text{valid committed} / \text{duration_seconds}$ TPS.
- Transaction success rate: $\text{valid committed} / \text{submitted} \times 100$.
- Average transaction size in bytes.
- Average block size in bytes.
- Average and maximum CPU utilization.
- Average and maximum memory utilization.
- Additional latency: $L_{\text{with_Fabric}} - L_{\text{without_Fabric}}$.
- Percentage latency increase: $(L_{\text{with_Fabric}} - L_{\text{without_Fabric}}) / L_{\text{without_Fabric}} \times 100$.
- CPU overhead: $\text{CPU}_{\text{with_Fabric}} - \text{CPU}_{\text{without_Fabric}}$.
- Memory overhead: $\text{Memory}_{\text{with_Fabric}} - \text{Memory}_{\text{without_Fabric}}$.

Required result

Use only valid transactions for the performance workload and retain submission, commit, block, transaction-size, and resource evidence.

Operator block

Not released. The valid-record load generator and commit/block-size collector must pass together before a command is added here.

R1 - Internet interruption and recovery

Design

- 3 repetitions.
- Each repetition is 30 min normal + 60 min proven external-Internet interruption + 30 min recovery.
- EMU-01 remains at the 15 s cadence for the full 120 minutes.
- The intended fault is external-Internet loss while the local system remains powered and observable.

Measure for normal, interruption, and recovery phases separately

- Local service availability.
- Delivery rate: $\text{successfully received/stored legitimate readings} / \text{transmitted readings} \times 100$.
- Data-loss count.
- Duplicate-record count.

- Database-blockchain consistency rate.
- End-to-end latency.
- Chronological-order accuracy using sequence/frame-counter/timestamp evidence.
- Recovery time from restoration until normal external connectivity returns.
- Mean and sample SD for latency and recovery measurements where applicable.

Required result

Essential local monitoring continues during the interruption and normal connectivity returns without database-blockchain inconsistency. Do not report queued or pending Fabric work as a committed blockchain record.

Operator block

Not released. The exact external-Internet outage/restore action must be rehearsed and controlled by one complete harness first.

R2 - Fabric unavailability and reconciliation

Design

1. 10 normal records reach valid confirmed Fabric state.
2. While Fabric is unavailable, create 10 additional valid records; these remain locally stored and pending, with no false commit.
3. Restore Fabric and reconcile those **same 10** pending records.

Measure

- Pending-record identification rate: $\text{correctly pending outage records} / 10 \times 100$.
- Recovery success rate: $\text{pending records successfully reconciled} / \text{pending records requiring reconciliation} \times 100$.
- Post-recovery hash-consistency rate: $\text{matching recovered DB/Fabric hashes} / \text{recovered verified records} \times 100$.
- False-verification count.
- Orphan database-record count.
- Missing blockchain-record count.
- Duplicate blockchain-record count.
- Conflicting-hash count.
- Failed-reconciliation count.
- Remaining-pending count.
- Recovery time from Fabric restoration until all recoverable records reach a final state.

Required result

100% pending identification, 0 false verification, 100% recovery of valid pending records, 100% post-recovery hash consistency, and 0 missing or duplicate valid blockchain anchors.

Operator block

Not released. Query-before-resubmission, exact-ten record freezing, and same-record reconciliation must be one proven harness first.

A1 - Authentication and access control

Design: 90 attempts

LoRaWAN, 10 trials each:

- Correct registered OTAA credentials -> allow.
- Registered identity with incorrect AppKey -> reject.
- Unregistered DevEUI -> reject/not activated.

MQTT, 10 trials each:

- Authorized account to allowed test topic -> accept.
- Incorrect password -> reject connection.
- Limited account to prohibited topic -> deny publish.

Fabric, 10 trials each:

- Authorized writer -> valid commit.
- Valid identity without writer permission -> reject; world state unchanged.
- Invalid/untrusted identity -> reject; world state unchanged.

Measure

- Authorized success rate.
- Unauthorized rejection rate.
- False-acceptance count.
- False-rejection count.
- Correct-decision rate.
- Response time by layer with mean and sample SD.
- Unauthorized state-change count.

Required result

0 false acceptance and 0 unauthorized state change. A LoRaWAN rejection counts only when gateway reception of the attempted request/frame is proven.

Operator block

Not released. The complete 9-condition x 10-trial fixture is not yet one commissioned action.

A2 - Fabric endorsement-policy enforcement

Design: 30 attempts

- 10 valid transactions with required endorsement available.
- 10 controlled missing-required-endorsement attempts.
- 10 valid transactions after endorsement availability is restored.

Measure

- Endorsement-policy enforcement rate.
- Normal valid-transaction success rate.
- Post-restoration success rate.
- Unauthorized state-change count.
- Response time by condition.

Required result

100% enforcement during policy violation, 0 unauthorized state changes, and valid commits resume after restoration.

Operator block

Not released. A Fabric-team-approved missing-endorsement fixture and world-state pre/post proof are required first.

S1 - LoRaWAN replay and spoofing

Design: 40 counted attempts

Replay:

- 10 legitimate controls.
- 10 old-frame replay attacks, each counted only after the gateway proves RF reception.

Spoofing:

- 10 genuine controls.
- 10 forged/invalid-MIC attacks, each counted only after the gateway proves RF reception.

Measure

- Replay-rejection rate.
- Spoofing-rejection rate.
- False-acceptance count.
- False-rejection count.
- Unauthorized-propagation rate into application/database/Fabric.
- Gateway-reception-to-decision time when the timestamps permit it.

Required result

All legitimate controls are processed; all received attack frames are rejected before application processing; 0 false acceptance; 0 attack-generated valid database/Fabric record.

Operator block

Not released. Capture, attack transmission, gateway-reception proof, and downstream absence checks must become one complete harness first.

S2 - Application-layer duplicate/replay

Chapter 3 lists this test but does not define a standalone trial count, duration, exact injection boundary, metric formula, or acceptance rule.

Status

METHODOLOGY REQUIRED. Do not invent counted results and do not substitute I3 or ordinary database deduplication. The methodology must first define the duplicate identifier, control/attack counts, tested layer, expected downstream behavior, state-change rule, and metric numerator/denominator.

F1 - MQTT invalid-connection flooding

Design

- 0 invalid connections/s, 10/s, and 50/s.

- 5 minutes per condition.
- 3 repetitions per condition = 9 load runs.
- 5-minute recovery after every run.
- Legitimate EMU-01 traffic continues at 15 s cadence.
- Flood only the isolated test listener, never the production gateway listener.

Measure

- Invalid-connection rejection rate.
- Legitimate-message delivery rate.
- Legitimate end-to-end latency.
- Valid-message throughput.
- Average and maximum CPU.
- Average and maximum memory.
- RX and TX bandwidth in Mbps.
- Service availability.
- Unauthorized database/Fabric record count.
- Recovery time after load stops.

Required result

Invalid traffic is rejected, legitimate traffic continues, no invalid request creates valid state, and no sustained service failure remains after recovery.

Operator block

Not released. The isolated-listener lifecycle and full 0/10/50 s⁻¹ orchestration must pass as one harness first.

F2 - Invalid application-message flooding

Design

- 0 invalid messages/s, 10/s, and 50/s.
- 5 minutes per condition.
- 3 repetitions per condition = 9 load runs.
- 5-minute recovery after every run.
- Legitimate EMU-01 traffic continues at 15 s cadence.
- Malformed data is published only to the isolated validation test path.

Measure

- Invalid-message rejection rate.
- Legitimate-message delivery rate.
- Legitimate end-to-end latency.
- Valid-message throughput.
- Average and maximum CPU.
- Average and maximum memory.
- RX and TX bandwidth in Mbps.
- Service availability.
- Unauthorized database/Fabric record count.
- Recovery time after load stops.

Required result

Malformed test messages never become valid database/Fabric records, legitimate traffic continues, and the service returns to the normal range after recovery.

Operator block

Not released. The isolated validation branch and full 0/10/50 s⁻¹ orchestration must pass as one harness first.

I1 - Application-layer alteration

Design: 20 attempts

- 10 unchanged controls.
- 10 deliberately altered records after the initial experimental hash and before valid storage.

Measure

- Integrity-verification rate.
- Tamper-detection rate.
- False-positive count.
- False-negative count.
- Unauthorized-storage count.
- Verification time with mean and sample SD.

Required result

All unchanged controls match and pass; all altered records mismatch and are blocked/quarantined; 0 false negatives; 0 altered records stored/submitted as valid.

Operator block

Not released. The controlled post-hash/pre-storage alteration fixture must be reproducible and reversible first.

I2 - Post-storage database tampering

Design: 20 attempts

- 10 unchanged stored controls with confirmed original Fabric evidence.
- 10 controlled post-storage value changes; restore each record after its trial.

Measure

- Integrity-verification rate.
- Tamper-detection rate.
- False-positive count.
- False-negative count.
- Unauthorized-storage/state count as applicable.
- Verification time with mean and sample SD.

Required result

All unchanged records match the original evidence; every modified record recomputes to a different digest using the same production canonicalization rules; 0 false negatives.

Operator block

Not released. A reviewed read-only current-row canonical verifier plus controlled tamper/restore fixture is required first.

I3 - Blockchain duplicate and overwrite integrity

Design

- Create and read back 10 valid baseline anchors.
- Submit 10 same-ID/same-hash duplicate attempts.
- Submit 10 same-ID/different-hash overwrite attempts.

Measure

- Duplicate-rejection rate.
- Overwrite-rejection rate.
- Original-hash preservation rate.
- Unauthorized state-change count.

Required result

Duplicate creates and conflicting overwrites are rejected and every original baseline hash remains unchanged.

Operator block

Not released. A dedicated synthetic mutation namespace and approved submit/query helpers are required first.

T1 - Individual record traceability

Design

- 10 known readings.
- Each reading is reconstructed from original source/LoRaWAN evidence through the database and its corresponding Fabric evidence.

Required trace fields

Trace identifier, Device EUI, LoRaWAN frame counter, source sequence, sensor value and unit, event timestamp, gateway identifier, database record identity, SHA-256 digest, Fabric transaction identifier, and commit timestamp or block reference.

Measure

- Trace-retrieval success rate.
- Record-completeness rate.
- Database-Fabric linkage rate.
- Missing-record count.
- Duplicate-record count.
- Individual retrieval time with mean and sample SD.

Required result

Complete retrieval and correct database-Fabric linkage for all tested records, with no expected record missing or duplicated.

Operator block

Not released. Full ten-record automatic completeness/linkage scoring must be commissioned first.

T2 - Device-history reconstruction

Design

- 10 non-overlapping sequences.
- 5 consecutive readings per sequence = 50 records.
- Together with T1, traceability covers 20 trials and 60 records.

Measure

- Record-completeness rate.
- Database-Fabric linkage rate.
- Chronological-order accuracy.
- Missing-record count.
- Duplicate-record count.
- History reconstruction time with mean and sample SD.

Required result

All five designated readings in every sequence are retrieved, correctly linked, and in the original source order; no expected record is substituted with a nearby reading.

Operator block

Not released. Automatic 10 x 5 source-to-database-to-Fabric reconstruction/scoring must be commissioned first.

Completion checklist

- P1: 3 valid 30-minute runs.
- P2: 12 valid with-Fabric workload runs plus the frozen no-Fabric comparison design.
- R1: 3 valid 120-minute runs with proven 30/60/30 phase boundaries.
- R2: 10 normal + 10 outage records + reconciliation of those same 10.
- A1: 90 counted attempts.
- A2: 30 counted attempts.
- S1: 40 counted attempts; every attack attempt must have proven gateway reception.
- S2: methodology must be formally resolved before counted work.
- F1/F2: 18 five-minute load runs total plus 18 five-minute recovery observations.
- I1/I2: 40 attempts total.
- I3: 10 baseline anchors + 20 mutation attempts.
- T1/T2: 20 trials covering 60 records.
- Every final reported value must be reproducible from retained raw evidence.